Project - Individual submission.

My approach is to use multiclass logistic regression (softmax with cross-entropy) to predict a player's poker decision.

This model can predict one of three decision (fold, call/check or raise)

Goal of this model is to estimate the probality of each action given the situation

Feature

Each decision is described by 5+9 feature

Starting hand strength considered.

starting_hand = a single index containing high & low

high_card_rank - the rank of the higher card

is_pair - 1 if both hole card has same rank

is_suited - 1 is both hole card has same suit

Betting Situation considered

Amount_to_Call - the amount the player must put

pot_size - the chips in pot

raises_so_far - the number of raises before this decision.

stack_bb - the players remaining stack

Table position: position_id - players seat

Method

Each decision n is described by a feature vector x_n

Its label is one-hot vector t_n

Each class k gets a score from feature $a_{nk} = \sum w_k^T x_n$

Softmax to turn the 3 scores into probabilities

Then the probabilities of observed action gives likelihood and negative log gives cross entropy loss

Then gradient is calculate with respect to weight and the weight is continuosly updated until the loss stops decreasing.